

K-Space: An Axiomatic Framework for Observer-Relative Measurement Registration in Quantum Theory

Viet Nguyen Xuan

viet@vvvqmrf.com

Draft v0.1 2026-06-02

Preprint anchor: [DOI: 10.5281/zenodo.20493403](https://doi.org/10.5281/zenodo.20493403) (CC BY 4.0)

Abstract

We introduce K-space, a formal structure defined by eight axioms (K1–K8) representing the content of observer-relative measurement registrations in quantum theory. K-space is not a physical space; it is a bookkeeping structure for what an observer has registered from quantum measurements, organized as a temporally ordered chain of K-state tuples, each carrying a registered outcome, a self-certification marker, a timestamp, and a validity status. The axioms define: the carrier set with admission and injectivity conditions (K1); a discrete temporal order (K2); intrinsic self-certification (K3); default validity with a null-event guard (K4); cross-observer invalidation dynamics (K5); cross-registration authority (K6); process closure transitioning provisional to final validity (K7); and cross-space embedding preservation (K8). A postulate K9_E extends K-space to a probability assignment $P(o \mid k_i, K_{\text{ctx}}) = \text{Tr}(E_o \rho)[1 - \beta f_\perp]/Z_E$, where $\beta \in [0, 1]$; setting $\beta = 0$ recovers the Born rule as a corollary. Three bridge theorems (T1, T9, T5) connect the axioms to operational constructions in multi-observer scenarios; the supporting Level-4 admissibility conditions, the incommensurability and EWF bridges (T2, T3), the colimit existence theorem (T4-H), and the associativity proof (T5) are developed self-containedly in the appendices. The framework is interpretively neutral and requires no prior familiarity with Buddhist philosophy of mind or other motivational contexts.

1 Introduction

A recurring difficulty in quantum foundations is specifying, in formal terms, what it means for a particular observer to *have* a measurement outcome. Standard quantum mechanics is silent on this: the Born rule assigns probabilities to outcomes, and the Hilbert space formalism evolves states, but neither provides a structure-theoretic account of the content of observer-relative measurement registrations. Approaches that center on the observer — Everett’s relative-state formulation [1], Wigner’s Friend gedankenexperiments [2], and extended multi-observer scenarios [3, 4, 5] — expose the need for a precise bookkeeping structure for “what one observer has registered and how that registration relates to what another has registered.”

Existing observer-relative frameworks do not fully supply this structure. Quantum Bayesianism (QBism) [6] locates quantum states in an agent’s degrees of belief but does not axiomatize a registration-content object or a validity lifecycle for that content. Relational quantum mechanics [7] makes all properties observer-relative but leaves the internal structure of “what the observer has” informal. Consistent histories [8] assigns probabilities to sequences of projections at the ensemble level, not to individual observer registration events. The many-worlds interpretation [1] identifies branches through decoherence but provides no formal bookkeeping for the specific content that one branch-observer has registered versus another. The Spekkens toy model [9] provides axiomatic epistemic states, but these track knowledge of preparation procedures over ontic states, not time-stamped registration events with a validity lifecycle.

This paper introduces K-space to fill exactly this gap. A K-space K_R for registering system R is a set of tuples $k = \langle M, o, \text{cert}, t, V \rangle$ representing individual registration events, equipped with a temporal chain order (K2), intrinsic certification (K3), default validity with invalidation dynamics (K4–K6), and a process-closure mechanism (K7). K-space is not a Hilbert space, not a phase space, and not a probability space; it is a registration-logic structure whose primitive predicates (cert , V , $\perp$, Auth) are epistemological rather than physical.

We also introduce postulate K9_E, which generalizes the Born rule by inserting a structural correction factor $[1 - \beta f_\perp]$ derived from the K-side context. At $\beta = 0$, K9_E reduces to the Born rule by corollary. At $\beta \neq 0$, K9_E produces measurable deviations in extended Wigner’s Friend experiments; the empirical determination of β is addressed in a companion paper [10].

Paper structure. Section 2 states and motivates the eight axioms K1–K8. Section 3 introduces K9_E and derives the Born rule as a corollary. Section 4 presents three bridge theorems (T1, T9, T5). Section 5 compares K-space with existing frameworks. Section 6 states three open problems. Section 7 concludes. Appendices A.1–E make the paper self-contained: T8 and the Born corollary (A.1–A.2), restricted φ_R existence (A.3), the

Level-4 admissibility definitions (B), the supporting bridges T2/T3/AJVS (C), the colimit existence theorem T4-H (D), and the associativity proof T5 (E).

2 K-Space Axioms K1–K8

Notation. $\mathcal{M}_K$ denotes the set of measurement-registration act identifiers (tokens, not types: two events of the same act type at different times are distinct tokens). O is the outcome alphabet; $\emptyset \in O$ is reserved for null events (zero information transfer). T_R is the discrete registration-time index set of R . $\Delta I(k)$ denotes the *information-transfer delta* of registration event k : the number of distinct field-values contributed by k to K_R beyond what a null token would contribute; $\Delta I(k) = 0$ iff k delivers no new information to K_R . Two K-state tuples k_1, k_2 are *K-incommensurable*, written $k_1 \perp_K k_2$, iff: (i) they fall within the same comparison context C_K (a Level-4 structural input supplied by the experimental protocol; defined in Appendix B, see also §6.2), and (ii) $o(k_1) \neq o(k_2)$ (conflicting registered outcomes). Equivalently, K5(ii) fires: $k_2 \perp k_1$ within C_K . For K-spaces: $K_{R_1} \perp_K K_{R_2}$ iff $\exists k_1 \in K_{R_1}, k_2 \in K_{R_2}$ such that $k_1 \perp_K k_2$.

Scope Note

Axioms K1–K8 define K-space as a formal object. They do *not* derive quantum mechanics from K-space, nor establish the existence of a full structure-preserving map $\varphi: K \rightarrow \mathcal{B}(\mathcal{H})$.

Partial result. A restricted map $\varphi_R: K_R \rightarrow \mathcal{P}(\mathcal{H}) \cup \{0\}$ has been shown to exist for the $N = 2$ extended Wigner’s Friend case (Class C theorem, 2026-06-01; Appendix A.3). This partial result does not imply the full φ -conjecture.

Open conjecture. The full map $\varphi: K \rightarrow \mathcal{B}(\mathcal{H})$ remains an open problem (§6.1).

Axiom 2.1 (Carrier Set — K1). *The K-side registration space of a registering system R is a set K_R whose elements are K-state tuples:*

$$K_R = \{ k \mid k = \langle M, o, \text{cert}, t, V \rangle \}$$

with $M \in \mathcal{M}_K$, $o \in O \cup \{\emptyset\}$, $\text{cert} \in \{0, 1\}$, $t \in T_R$, $V \in \{0, 1\}$.

Admission rule: $k \in K_R \Rightarrow \text{cert}(k) = 1$ (only self-certified events are admitted to K_R).

Injectivity: $t \upharpoonright K_R$ is injective: $\forall k_1, k_2 \in K_R, t(k_1) = t(k_2) \Rightarrow k_1 = k_2$.

Claim class: C (conjectural VVV-QMRF formal definition).

Remark 2.1. K_R records *what was registered*, not what physically exists. Elements k are not density matrices, not wave functions, and not probability values. The slot $o = \emptyset$ is reserved for null events (K4). K1 carries claim class C because the carrier-set definition is,

as a whole, a conjectural formal claim about the existence of a well-formed registration space; K2–K8 carry class D as proposed structural operations acting on an already-admitted carrier. Among the five fields, M, o, cert, t are the immutable record of the event; V is the only *dynamic* field. The value V listed in the tuple is the validity status *at instantiation*; its subsequent value is the time-indexed status $V_{\text{prov}}(k)$ (provisional, pre-closure) or $V_{\text{final}}(k)$ (final, post-closure), governed by K4–K5 and K7. References to $V(k)$ in K5–K8 denote $V_{\text{prov}}(k)$ before K7 closure and $V_{\text{final}}(k)$ after.

Axiom 2.2 (Temporal Order — K2). $(K_R, <_R)$ is a strict total order (chain) where $k_1 <_R k_2 \iff t(k_1) < t(k_2)$. The order is discrete: for any consecutive pair $k_i, k_{i+1} \in K_R$ with no element strictly between them, no registration state exists in the open interval $(t(k_i), t(k_{i+1}))$ on the K-side.

Claim class: D (proposed).

Remark 2.2. K2 does not claim physical time is discrete; only the K-side registration-time index within K_R is discrete. Physical time in Hilbert space remains continuous.

Axiom 2.3 (Self-Certification — K3). $\sigma_R: \mathcal{M}_K \rightarrow \{0, 1\}$ with

$$\sigma_R(M) = 1 \iff M \text{ has occurred as a K-side registration event of } R,$$

determined intrinsically within K_R . For all $k \in K_R$: $\text{cert}(k) = \sigma_R(M(k))$.

Observer-indexed independence: $\sigma_R(M)$ is independent of $\sigma_{R'}(M')$ for any $R' \neq R$.

Claim class: D.

Remark 2.3. $\sigma_R(M)$ certifies the *occurrence* of M as a K-side event. It does not certify that the physical outcome is “correct,” is not consciousness, and is not a second-order measurement act.

Axiom 2.4 (Default Validity — K4). Define $\text{isNull}(k) := o(k) = \emptyset \wedge \Delta I(k) = 0$ (zero-information-transfer event). For all $k \in K_R$:

- (a) $\neg \text{isNull}(k) \Rightarrow V(k) = 1$ upon instantiation,
- (b) $\text{isNull}(k) \Rightarrow V(k) = 0$.

Claim class: D.

Remark 2.4. $V(k) = 1$ is the default K-side validity for non-null events: it means this registration is treated as valid within K_R until contradicted (K5). It does not mean the physical outcome is correct. Validity is provisional until process closure (K7).

Axiom 2.5 (Invalidation — K5). K5 fires only when a comparison context C_K exists (i.e., when joint registration is structurally demanded). $V(k_1) \rightarrow 0$ iff $\exists k_2$ in the relevant K-space such that:

- (i) $k_1 <_R k_2$ (K2: k_2 is temporally later),
- (ii) $k_2 \perp k_1$ within shared C_K (registered contradiction),
- (iii) $\text{Auth}(k_2 \rightarrow k_1, C_K) = 1$ (K6: cross-registration authority).

K5 does not fire when no C_K exists. Before process closure (K7), $V(k_1) = V_{\text{prov}}(k_1)$ is provisional and in principle reversible (if k_2 is itself invalidated before closure). After K7 closure, $V_{\text{final}}(k_1) = 0$ is absolute and irreversible.

Claim class: D.

K5_prospective [EXTENSION — required for K9_E only]

For probability assignment via K9_E (§3), K5 admits a *prospective evaluation mode* on hypothetical tuples $k_o^* = \langle M^*, o, 1, t^*, 1 \rangle$ representing a registration that *would* be instantiated if outcome o were realized.

K5 fires prospectively on k_o^* iff requires $K_{\text{joint}} = 1$ and $\exists k_{\text{prev}} \in K_{\text{joint}}$ such that: (i) $k_{\text{prev}} <_{\text{joint}} k_o^*$, (ii) $k_o^* \perp k_{\text{prev}}$ within C_K , (iii) $\text{Auth}(k_o^* \rightarrow k_{\text{prev}}, C_K) = 1$. Conditions (i)–(iii) are identical to K5; only the evaluation target differs. Prospective firing does *not* modify any V in K_R ; it contributes only to $f_{\perp}(o)$ in K9_E (Theorem T8, Appendix A.1).

Axiom 2.6 (Cross-Registration Authority — K6). $\text{Auth}(k_2 \rightarrow k_1, C_K) = 1$ iff all of:

- (a) $C_K\text{-sphere}(k_1) = C_K\text{-sphere}(k_2)$ (same comparison context),
- (b) $V(k_2) = 1$ (k_2 not invalidated at check time),
- (c) $k_1 \in \text{scope}(D_{\text{joint}})$ (k_1 's claim is within the joint-validity demand).

Authority is non-hierarchical and non-transitive across distinct contexts. Two observers in the same C_K may hold mutual authority when both are valid.

Claim class: D.

Axiom 2.7 (Registration Process Closure — K7). R closes at $t_{\text{close}}(K_R)$ iff all pending joint-validity demands involving K_R are resolved. At closure, for all $k \in K_R$: $V_{\text{prov}}(k) \rightarrow V_{\text{final}}(k)$. Post-closure properties:

- (a) No new k can be instantiated in K_R ,
- (b) K5 irreversibility is absolute ($V_{\text{final}}(k) = 0$ is permanent),
- (c) No new joint-validity demand D_{joint} involving K_R can be raised,
- (d) K_{joint} involving K_R is final.

Claim class: D.

Axiom 2.8 (Cross-Space Embedding Preservation — K8). For any embedding $i: K_R \rightarrow K_X$ (where K_X is a joint or cross-space structure):

- (i) $V_X(i(k)) = V_R(k)$ at embedding time t_{embed} for all $k \in K_R$ (validity snapshot preserved),
- (ii) All five tuple fields $(M, o, \text{cert}, t, V)$ are preserved exactly under i ,

(iii) After embedding, $V_X(i(k))$ evolves by K4–K7 rules within K_X ; K8 does not immunize against future K5 invalidation in K_X .

K8 is independent of K4: K4 governs validity at native instantiation; K8 governs validity-preservation at cross-space transfer.

Claim class: D.

Table 1 summarizes the layer-1 axiom cluster.

Table 1: K-space axioms K1–K8: roles, field coverage, and claim class. Class C = structurally testable conjecture; Class D = proposed. Neither modifies standard QM. “Lvl-4” indicates a conditional semantic dependency on Level-4 structural definitions (comparison context C_K , joint-validity demand D_{joint} , predicate requires_K_joint).

Axiom	Name	Fields	Class	Lvl-4 dep.
K1	Carrier set	M, o, cert, t, V	C	None
K2	Temporal order	t	D	None
K3	Self-cert.	cert	D	None
K4	Default validity	V (default)	D	None
K5	Invalidation	V (transition)	D	C_K
K6	Authority	V (auth. cond.)	D	$C_K, \text{scope}(D_j)$
K7	Closure	V (closure)	D	req. $_{KJ}$
K8	Embedding pres.	all five	D	None

3 K9_E: Probability Postulate

Axioms K1–K8 define K-space as a structural object but do not assign probabilities. Postulate K9_E supplies the probability assignment.

Definition 3.1 (K-context set). For registering system R_i performing experiment Exp, the *contextual K-state set* is

$$K_{\text{ctx}}(k_i, \text{Exp}) = \left\{ \varphi_{ij}(k_j) \in K_{\text{joint}} \mid k_j \in K_{R_j}, \right. \\ \left. R_j \in \text{Obs}(\text{Exp}, R_i), k_j \text{ temporally compatible with } k_i \right\} \quad (1)$$

where φ_{ij} is the K8-constrained canonical embedding constructed in Theorem T9 (§4) below; its existence and uniqueness are established there without additional assumption (beyond K1–K8 + T1). The (finite, R_i -relative) *observer set* is

$$\text{Obs}(\text{Exp}, R_i) := \{ R_j \neq R_i : \text{requires_K_joint}(R_i, R_j) = 1 \text{ and } R_j \text{ participates in Exp} \};$$

this membership criterion (definition D_obs) replaces the earlier informal “other observer” clause and makes K_{ctx} fully formal (see Appendix B for the underlying Level-4 predicates).

When $\text{Obs}(\text{Exp}, R_i) = \emptyset$ one has $K_{\text{ctx}} = \emptyset$ and K9_E reduces to the Born rule (cf. the empty-context convention in Def. 3.2).

Definition 3.2 (Perpendicularity fraction).

$$f_{\perp}(o, K_{\text{ctx}}) = \frac{|\{k_j \in K_{\text{ctx}} : k_o^* \perp_K k_j\}|}{|K_{\text{ctx}}|} \quad (2)$$

where $k_o^* = \langle M^*, o, 1, t^*, 1 \rangle$ is the hypothetical outcome- o tuple (cf. the K5_prospective box, §2.5). Since all $k_j \in K_{\text{ctx}}$ share the comparison context, $k_o^* \perp_K k_j \iff o(k_j) \neq o$; thus $f_{\perp}(o, \cdot)$ counts the fraction of context registrations whose outcome *conflicts* with the candidate o , and is therefore genuinely o -dependent — the property that makes the $[1 - \beta f_{\perp}]$ factor survive normalization (cf. alternative A2 in Theorem T8). Equivalently by Theorem T8 (Appendix A.1):

$$f_{\perp}(o, K_{\text{ctx}}) = \mathbb{E}[\mathbf{1}(\text{K5_prospective fires on } k_o^* \text{ vs } k_j \in K_{\text{ctx}})].$$

Empty-context convention: when $|K_{\text{ctx}}| = 0$ (no other registering system satisfies requires_K_joint), set $f_{\perp}(o, K_{\text{ctx}}) := 0$. By Eq. (3) this collapses K9_E to the Born rule, consistent with Corollary 3.0.1: an isolated observer incurs no K-side suppression.

Postulate 3.1 (K9_E Probability Postulate).

$$\boxed{P(o \mid k_i, K_{\text{ctx}}) = \frac{\text{Tr}(E_o \rho) [1 - \beta f_{\perp}(o, K_{\text{ctx}})]}{Z_E}} \quad (3)$$

where E_o is the POVM element for outcome o , ρ is the quantum state, $\beta \in [0, 1]$ is a free parameter, and $Z_E = \sum_{o'} \text{Tr}(E_{o'} \rho) [1 - \beta f_{\perp}(o', K_{\text{ctx}})]$ ensures $\sum_o P(o \mid k_i, K_{\text{ctx}}) = 1$.

Well-posedness ($Z_E > 0$). $Z_E > 0$ holds whenever there exists an outcome o' with $\text{Tr}(E_{o'} \rho) > 0$ and $\beta f_{\perp}(o', K_{\text{ctx}}) < 1$ (in particular for any $\beta < 1$, since $f_{\perp} \leq 1$ and $\sum_{o'} \text{Tr}(E_{o'} \rho) = 1$). The sole degenerate case $Z_E = 0$ requires $\beta = 1$ together with $f_{\perp}(o', K_{\text{ctx}}) = 1$ for every outcome o' carrying nonzero Born weight; in this measure-zero boundary configuration K9_E is left undefined and the assignment defaults to the Born rule by the convention of Corollary 3.0.1.

Corollary 3.0.1 (Born Rule Recovery). At $\beta = 0$, K9_E recovers the Born rule exactly: $P(o \mid k_i, K_{\text{ctx}})|_{\beta=0} = \text{Tr}(E_o \rho)$.

Proof. At $\beta = 0$: $Z_E = \sum_{o'} \text{Tr}(E_{o'} \rho) = \text{Tr}(\mathbf{1} \rho) = 1$ by POVM completeness. Therefore $P(o) = \text{Tr}(E_o \rho)$. $\square$

Remark 3.1. K9_E is a generalization of the Born rule parameterized by β . At $\beta = 0$, K-space is present but produces no deviation from standard QM predictions. The empirical

determination of whether $\beta \neq 0$ in nature is the scope of a companion paper [10], which proposes a modified Bong et al. protocol sensitive to K-side suppression effects.

4 Bridge Theorems

We present three bridge theorems connecting K1–K8 to operational constructions in multi-observer scenarios. The labels T1, T9, T5 follow the framework’s internal numbering; the supporting results (Level-4 admissibility conditions, the incommensurability theorem T2, the EWF bridge T3, and the colimit existence theorem T4-H) are developed self-containedly in Appendices B–E. The three theorems are presented here in operational dependency order ($T1 \rightarrow T9 \rightarrow T5$) rather than index order.

4.1 T1: K_{joint} Construction

Theorem 4.1 (Joint K-Space Construction — T1). *Given registering systems A, B with $\text{requires_K_joint}(A, B) = 1$ via a shared joint-validity demand D_{joint} , there exists a candidate joint K-space $K_{\text{joint}}(A, B)$ as the categorical colimit of the embedding diagram (K_A, K_B) with K1–K8-preserving morphisms $i_A: K_A \rightarrow K_{\text{joint}}$ and $i_B: K_B \rightarrow K_{\text{joint}}$.*

The combined order is the transitive closure

$$<_{\text{joint}} = (i_A(<_A) \cup i_B(<_B) \cup \text{cross-rel})^+,$$

where cross-rel encodes cross-structure temporal relations from the shared laboratory history. $(K_{\text{joint}}, <_{\text{joint}})$ is a partial order; restricted to each $i_X(K_X)$ it is a chain (from K2).

Remark 4.1. T1 is a *composition theorem*, not a pure K1–K8 derivation. It combines K1–K8 with external Level-4 inputs (cross-structure temporal relations from laboratory protocol). K1–K8 alone do not supply these relations. Existence of a candidate K_{joint} does not guarantee admissibility; the full admissibility conditions AdmJoint (i)–(v) are given in Appendix B.

4.2 T9: K_{ctx} Construction

Theorem 4.2 (K_{ctx} as Theorem — T9). *When $\text{requires_K_joint}(R_i, R_j) = 1$, define $\varphi_{ij}: K_{R_j} \rightarrow K_{\text{joint}}$ by $\varphi_{ij}(k_j) = i_j(k_j)$, where i_j is the K8-constrained canonical embedding from T1. Then φ_{ij} is the unique morphism preserving all five tuple fields, and K_{ctx} (Eq. (1)) is a theorem construction from K1–K8 + T1 with no additional assumption.*

Proof (four lemmas). **L1 (Existence).** $\text{requires_K_joint}(R_i, R_j) = 1 \Rightarrow$ T1 constructs K_{joint} with canonical embedding i_j . Define $\varphi_{ij} = i_j$; existence follows.

L2 (Uniqueness). Let $\psi: K_{R_j} \rightarrow K_{\text{joint}}$ satisfy K8 field preservation. For each k_j , $\psi(k_j)$

must have the same five fields as k_j (K8). By K1 t-injectivity, K_{joint} contains exactly one element with those fields: $i_j(k_j)$. Hence $\psi = \varphi_{ij}$; the morphism is structurally forced.

L3 (Field sufficiency). K8 preserves M, o, cert, t, V , which are exactly the fields required by K5_prospective (outcome o , temporal t , authority V , admission cert) and by K6 (validity V).

L4 (Exhaustion). Alternative channels are eliminated: (A) direct cross-space comparison without K_{joint} is undefined (no C_K); (B) ρ -side correlation is a category error (K-side requires tuple fields); (C) non-K8-compliant embeddings violate K8; (D) any other K8-compliant embedding equals φ_{ij} by L2. $\square$

Remark 4.2 (Historical note: elimination of assumption [A-E1]). Earlier drafts of T9 required an explicit assumption [A-E1]: “a T3-morphism $K_{R_j} \rightarrow K_{\text{joint}}$ exists.” Lemmas L1–L3 above show this assumption is redundant: the K8-constrained canonical embedding φ_{ij} exists (L1), is unique (L2), and supplies all fields required by K5_prospective and K6 (L3). Assumption [A-E1] is therefore eliminated; no assumption beyond K1–K8 + T1 is required. The supporting bridge theorems T2–T3 (Appendix C) record the general morphism and incommensurability theory from which [A-E1] was originally drawn.

4.3 T5: Associativity (Conditional)

Theorem 4.3 (K_{joint} Associativity — T5, conditional). *Given admissible K-side spaces K_A, K_B, K_C :*

$$K_{\text{joint}}(K_{\text{joint}}(A, B), C) \cong K_{\text{joint}}(A, B, C)$$

as K1–K8-structured sets. This is conditional on: (C1) the N-observer colimit existence theorem T4-H (Full Theorem, 4/4 steps verified 2026-05-28); and (C2) the global commutativity condition F7d for the $N = 3$ embedding diagram (confirmed via T4-H Step 3 content-basedness of K5).

Remark 4.3 (T5: proof in Appendix E). A self-contained proof of T5 is given in Appendix E: the colimit existence theorem T4-H (Appendix D) supplies the universal property, and the content-basedness of K5 firing in the colimit (Appendix E, Lemma B2) establishes the global commutativity condition F7d for the $N = 3$ embedding diagram.

Remark 4.4 (Bridge.EWF application (T3, Appendix C)). In an Extended Wigner’s Friend (EWF) configuration, Friend F registers definite outcome o_F and Wigner W registers the same laboratory in a coherent superposition. The EWF setup fixes the laboratory time order $t_F < t_W$ (so $k_F <_{\text{joint}} k_W$); the derivation presupposes this ordering and is not stated for $t_W < t_F$. Under candidate K_{joint} , K5 then forces a conflict: $k_W \perp k_F$ within C_K with $\text{Auth}(k_W \rightarrow k_F, C_K) = 1$, so $V(k_F) \rightarrow 0$ (at the admissibility level, at least one of $V(k_F), V(k_W)$ must drop), violating the admissibility condition. The firing *direction* is set by the laboratory time order, not by the axioms alone. This derivation additionally

requires the Axiom of Joint Validity Semantics (AJVS — a named semantic postulate requiring that K_{joint} hosts original first-order K-side validity claims, not meta-descriptions of the form “within K_F , M_F registered $|h\rangle$ ”; see Appendix C for the full statement).

5 Relation to Existing Frameworks

Table 2 compares K-space with five existing frameworks.

Table 2: Structural comparison of K-space with observer-relative and epistemic quantum frameworks. CH: Consistent Histories. MWI: Many-Worlds. *Spekkens tracks epistemic states over ontic states, not time-stamped registration tuples with a validity lifecycle.

Property	K-space	QBism	RQM	CH	MWI	Spekkens
Observer-indexed	Yes	Yes	Yes	No	No (branch)	Yes
Formal axioms	K1–K8	No	Partial	Yes	Partial	Yes
Registration content	Yes	No	No	No	No	No*
Invalidation rule	K5	—	—	—	—	—
Closure property	K7	—	—	—	—	—
Born rule	K9_E ($\beta=0$)	Agent	Relat.	Std.	Derived	Analog

vs. QBism [6]. QBism centers on agent degrees of belief; K-space centers on formal registration content. K9_E is a structural postulate, not a subjective prior.

vs. Relational QM [7]. Both are observer-relative. K-space additionally axiomatizes the *content* of what is registered (K1–K4), an invalidation mechanism (K5–K6), and a closure property (K7). RQM leaves the registration structure informal.

vs. Consistent Histories [8]. CH assigns probabilities to projection sequences at the ensemble level. K-space operates at the level of individual observer registration events with a validity lifecycle.

vs. Many-Worlds [1]. Everett branching identifies branches through decoherence but provides no formal bookkeeping for the content of what one branch-observer has registered versus another. K-space supplies this via K1–K8 and the K_{joint} construction (T1).

vs. Spekkens toy model [9]. Spekkens tracks degrees of knowledge about preparation procedures (epistemic states over ontic states). K-space tracks registered *outcomes* as formal tuples $\langle M, o, \text{cert}, t, V \rangle$ with a validity lifecycle (K4→K5→K7). These are orthogonal formalization targets: preparation-epistemic vs. outcome-registration-formal.

Non-claim. K-space is not presented as superior to these frameworks. It provides a minimal formal complement: axioms for registration content that the other frameworks leave implicit.

6 Open Problems

6.1 The full φ -map conjecture

Conjecture. Does there exist a full structure-preserving map $\varphi: K \rightarrow \mathcal{B}(\mathcal{H})$?

Partial result. A restricted map $\varphi_R: K_R \rightarrow \mathcal{P}(\mathcal{H}) \cup \{0\}$ has been shown to exist for the $N = 2$ EWF case (Class C theorem, 2026-06-01; Appendix A.3). Necessary conditions N_1 – N_8 and N_T for the full map have been derived from K1–K8.

Why open. Extending the restricted existence result to the full K-space requires characterizing the image of φ in $\mathcal{B}(\mathcal{H})$ under all K1–K8 structure-preserving constraints — a question in the representation theory of registration-logic structures.

6.2 Observer set selection rule for K_{ctx}

Problem. K_{ctx} is formally constructed by T9 (Theorem 4.2), and the observer set $\text{Obs}(\text{Exp}, R_i)$ now has a formal membership criterion — $\text{requires_K_joint}(R_i, R_j) = 1$ together with participation in Exp (definition D_obs, stated with Eq. (1)). What remains open is not the *form* of Obs but its *extension*: the rule fixing *which* physical interactions set $\text{requires_K_joint} = 1$ is supplied by the experimental protocol (Level-4 D_{joint} scope), not by K1–K8 alone.

Why open. The predicate requires_K_joint encodes which physical interactions create mutual validity dependencies. A fully axiomatic treatment requires connecting the K-side registration layer to the experimental-design layer.

6.3 Full N-observer colimit chain

Problem. T4-H (N-observer colimit existence) is a Full Theorem (4/4 steps, 2026-05-28). T5 (composition associativity) is Class C conditional (on T4-H and global commutativity F7d). The full chain for $N > 3$ observers requires extending the F7d commutativity check to arbitrary observer diagrams.

Why open. Non-transitivity of K-side incommensurability ($K_A \perp_K K_B \wedge K_B \perp_K K_C \not\Rightarrow K_A \perp_K K_C$, by T4) means each pair in an N -observer configuration requires an independent check. For large N , the combinatorial structure of the $N(N-1)/2$ required checks may reveal structural constraints not visible at $N = 2, 3$.

7 Conclusion

We have introduced K-space, a formal structure defined by eight axioms (K1–K8) representing observer-relative measurement registrations in quantum theory. The axioms collectively define a carrier set with admission and injectivity conditions (K1), a discrete temporal chain order (K2), intrinsic self-certification (K3), default validity with a null-event guard (K4), cross-observer invalidation (K5), cross-registration authority (K6), process closure transitioning provisional to final validity (K7), and cross-space embedding preservation (K8). Three bridge theorems — T1 (K_{joint} construction), T9 (K_{ctx} as a derived theorem, eliminating a prior morphism assumption via five lemmas), and T5 (associativity, conditional) — connect K1–K8 to operational constructions in multi-observer scenarios.

Postulate K9_E generalizes the Born rule to $P(o \mid k_i, K_{\text{ctx}}) = \text{Tr}(E_o \rho)[1 - \beta f_{\perp}]/Z_E$, with $\beta = 0$ recovering the Born rule exactly (Corollary 3.0.1). The empirical determination of β is the subject of the companion paper [10].

Three open problems are identified with precise formulations: the full φ -map conjecture (with a partial restricted-existence result for the $N = 2$ EWF case), the observer set selection rule for K_{ctx} , and the full N -observer colimit chain. These problems define a research program.

Acknowledgments

The formal structure of K-space was developed within the VVV-QMRF research framework. The author thanks the quantum foundations community for discussions on observer-relative approaches to quantum measurement.

References

- [1] H. Everett, “Relative state” formulation of quantum mechanics, *Rev. Mod. Phys.* **29**, 454 (1957).
- [2] E. P. Wigner, Remarks on the mind-body question, in *The Scientist Speculates*, ed. I. J. Good (Heinemann, 1961).
- [3] D. Frauchiger and R. Renner, Quantum theory cannot consistently describe the use of itself, *Nat. Commun.* **9**, 3711 (2018).
- [4] K.-W. Bong *et al.*, A strong no-go theorem on the Wigner’s friend paradox, *Nat. Phys.* **16**, 1199–1205 (2020).

- [5] H. M. Wiseman, E. G. Cavalcanti, and E. G. Rieffel, A “thoughtful” local friendliness no-go theorem, *Quantum* **7**, 1112 (2023).
- [6] C. A. Fuchs, N. D. Mermin, and R. Schack, An introduction to QBism with an application to the locality of quantum mechanics, *Am. J. Phys.* **82**, 749 (2014).
- [7] C. Rovelli, Relational quantum mechanics, *Int. J. Theor. Phys.* **35**, 1637 (1996).
- [8] R. B. Griffiths, *Consistent Quantum Theory* (Cambridge University Press, 2002).
- [9] R. W. Spekkens, Evidence for the epistemic view of quantum states: A toy theory, *Phys. Rev. A* **75**, 032110 (2007).
- [10] V. Nguyen Xuan, Have optical Wigner’s friend experiments been blind to a geometric degree of freedom? Zenodo, DOI: [10.5281/zenodo.20506279](https://doi.org/10.5281/zenodo.20506279) (2026).

A Supplemental Proofs

A.1 Theorem T8: K5_prospective Frequency Bridge

Theorem A.1 (T8 — K5_prospective Frequency Bridge). *Under K1–K8 binary primitives and a uniform context set K_{ctx} ,*

$$f_{\perp}(o, K_{\text{ctx}}) = \mathbb{E}[\mathbf{1}(\text{K5_prospective fires on } k_o^* \text{ vs } k_j \in K_{\text{ctx}})].$$

The fraction form (2) is the unique functional form consistent with binary K-side primitives.

Proof. For each $k_j \in K_{\text{ctx}}$, define indicator $I_j(o) = 1$ iff K5_prospective fires on k_o^* vs k_j (equivalently $k_o^* \perp_K k_j$, i.e. $o(k_j) \neq o$ within the shared C_K); else $I_j(o) = 0$. Then:

$$\frac{1}{|K_{\text{ctx}}|} \sum_j I_j(o) = \frac{|\{k_j : k_o^* \perp_K k_j\}|}{|K_{\text{ctx}}|} = f_{\perp}(o, K_{\text{ctx}}).$$

Uniformity of weight ($w_j = 1$ for all j) is structurally forced: all K-side primitives ($\perp$, Auth, V , cert, temporal order) are binary-valued in $\{0, 1\}$ (K2–K6); no continuous contradiction-strength metric exists in K1–K8. Four natural alternative weightings are each independently eliminated by structural constraints: (A1) quantum-overlap weight is circular (uses Born-rule probabilities that K9_E is intended to modify); (A2) outcome-independent binary indicator collapses to $\delta P = 0$ under normalization; (A3) Auth-weighted form reduces to uniform because $\text{Auth} \in \{0, 1\}$ (K6); (A4) temporal-decay weight introduces a new free parameter beyond the β budget. The fraction form is the unique survivor. $\square$

A.2 Born Rule Corollary: Proof

Restated for completeness: at $\beta = 0$, $Z_E = \sum_{o'} \text{Tr}(E_{o'}\rho) = \text{Tr}(\mathbf{1}\rho) = 1$ by POVM completeness, so $P(o) = \text{Tr}(E_o\rho)$. $\square$

A.3 Restricted φ_R Existence (Summary)

Theorem A.2 (Restricted existence of φ_R). *Let K_R be a K -space satisfying K1–K8, with outcome set $O = \{o_1, \dots, o_m\}$ for measurement $M \in K_R$. Let $\mathcal{H}$ be a Hilbert space with $\dim \mathcal{H} \geq |O|$ and orthonormal basis $\{|o_i\rangle\}_{i=1}^m$. Define*

$$\varphi_R(k) := \begin{cases} |o\rangle\langle o| & \text{if } V(k) = 1 \text{ } (o = \text{outcome of } k), \\ 0_{\mathcal{B}(\mathcal{H})} & \text{if } V(k) = 0. \end{cases} \quad (4)$$

Then $\varphi_R: K_R \rightarrow \mathcal{P}(\mathcal{H}) \cup \{0\}$ is total and well-defined, and satisfies necessary conditions N_1 – N_8 and N_T derived from K1–K8 and Theorems T1–T3 (verified in Table 3; T2 and T3 are stated in Appendix C). For N_T : when $K_{R_1} \perp_K K_{R_2}$, the two registrations stem from incompatible measurements whose outcome projectors are non-orthogonal on the shared registration Hilbert space, so $[\iota(\varphi_R(k_1)), \varphi_R(k_2)] \neq 0$. (The orthonormal basis $\{|o_i\rangle\}$ indexes the outcomes of a single measurement M ; outcomes of distinct incommensurable measurements are not mutually orthogonal, which is precisely what makes the commutator nonzero.)

For $N \geq 3$ registering systems, a unique colimit extension $\varphi_{\text{colim}}: K_{\text{joint}}(R_1, \dots, R_N) \rightarrow \mathcal{B}(\mathcal{H}_{\text{joint}})$ exists via the universal property of T4-H (Full Theorem, 2026-05-28) and satisfies conditions φ -N1 (uniqueness), φ -N2 (associativity, conditional on T5), and φ -N3 (pair commutator). The $N \geq 3$ extension is Class C conditional: φ -N2 inherits the conditional status of T5 (§4, §6.3), whose proof is given in Appendix E. The $N = 2$ restricted existence result above is unconditional (Class C theorem).

Proof. Construction (4) is explicit and unambiguous (K1 admission guarantees well-definedness; K4 dichotomy covers all $k \in K_R$). Every $V(k) = 1$ event of M carries a genuine outcome $o \in O$, so $|o\rangle$ is a basis vector; the reserved null slot $o = \emptyset$ has $V = 0$ by K4(b) and maps to $0_{\mathcal{B}(\mathcal{H})}$, so φ_R is total on K_R . Table 3 verifies each condition. For $N \geq 3$: the compatible family $\{\iota_i \circ \varphi_i\}$ satisfies T4-H Step 4 universality (Appendix D); the unique mediating morphism φ_{colim} is set equal to it (T5 associativity: Appendix E). $\square$

Table 3: Verification of necessary conditions N_1 – N_8 and N_T for φ_R defined by Eq. (4).

Cond.	Source	Verification	OK
N_1 (well-def.)	K1	φ_R total; image closed under Lüders products	✓
N_2 (Lüders ord.)	K2	$t(k_1) < t(k_2) \Rightarrow P_{o_2} P_{o_1} P_{o_2}$ Lüders	✓
N_3 (cert-refl.)	K3	$P_o^2 = P_o = P_o^\dagger$; $\text{Im}(\varphi_R) \subseteq \text{Proj} \cup \{0\}$	✓
N_4 (valid-pos.)	K4	$V=1 \Rightarrow o\rangle\langle o \geq 0$; $\text{isNull} \Rightarrow \varphi_R=0$	✓
N_5 (inval.-abs.)	K5	$V \rightarrow 0$ absorbing: $\varphi_R=0_{\mathcal{B}(\mathcal{H})}$	✓
N_6 (auth.-comp.)	K6	$\text{Auth}=1 \Rightarrow P_{o_2} P_{o_1} \neq 0$ (necessary); sufficiency: boundary	✓*
N_7 (closure-fin.)	K7	φ_{final} fixed at t_{close} ; no post-closure change	✓
N_8 (embed.-nat.)	K8	$\varphi_X \circ i = \iota \circ \varphi_R$; $\iota(P_o) = P_o \otimes \mathbf{1}$	✓
N_T (K-incomm.)	T2/T3	$K_{R_1} \perp_K K_{R_2} \Rightarrow [\iota(\varphi_R(k_1)), \varphi_R(k_2)] \neq 0$ (non-orth. projectors)	✓

*Necessary direction holds; the converse (sufficiency, $P_{o_2} P_{o_1} \neq 0 \Rightarrow \text{Auth} = 1$) is the boundary of Theorem A (box below). N_T holds because incommensurable registrations map to non-orthogonal projectors on the shared registration Hilbert space (Appendix C, T2/T3).

Boundary of restricted existence

The map φ_R on $\mathcal{P}(\mathcal{H}) \cup \{0\}$ is proven. Two structural limits apply, with proof sketches below:

Theorem A (N_6 sufficiency boundary). The necessary direction $\text{Auth}(k_2 \rightarrow k_1, C_K) = 1 \Rightarrow P_{o_2} P_{o_1} \neq 0$ holds; the converse is *not provable* from $\mathcal{B}(\mathcal{H})$ alone. Sketch (4 steps): (1) $\mathcal{B}(\mathcal{H})$ encodes only operator-algebraic data (projectors, products, commutators, spectra), not epistemic-sphere membership or authority scope; (2) by K6, $\text{Auth} = 1$ requires C_K -sphere co-membership (a) and $k_1 \in \text{scope}(D_{\text{joint}})$ (c), both K-side predicates with no operator-algebraic definition; (3) two projectors with $P_{o_2} P_{o_1} \neq 0$ can lie in *different* C_K -spheres (independent experiments sharing $\mathcal{H}$), which $\mathcal{B}(\mathcal{H})$ cannot distinguish from co-membership; (4) hence no $\mathcal{B}(\mathcal{H})$ -only proof of the converse exists.

Theorem B (global commutativity boundary, $N > 2$). The pairwise conditions $K_{R_i} \perp_K K_{R_j} \Rightarrow [\iota_i(P_{o_i}), \iota_j(P_{o_j})] \neq 0$ hold, but global K_{joint} path-commutativity is *not* determined by them. Sketch (2 steps): (1) $K_{\text{joint}}(R_1, \dots, R_N) = \text{colim}(D)$ is fixed by a global universal property (Appendix D, Step 4), and path-commutativity is a global categorical property, not decomposable into pairwise relations without the coherence supplied by T5 (Appendix E); (2) the $N(N-1)/2$ pairwise commutators encode only binary relations, so no pairwise $\mathcal{B}(\mathcal{H})$ data recovers global colimit coherence.

These are characterized limits, not defects, and reflect current understanding rather than permanent impossibility: an operator-algebraic encoding of C_K -sphere membership would resolve Theorem A, and a $\mathcal{B}(\mathcal{H})$ -expressible global coherence condition would resolve Theorem B.

φ_R is a *correspondence map* with precisely characterized limits. $\text{Im}(\varphi_R) = \{|o\rangle\langle o|\} \cup \{0\}$ is the tightest $\mathcal{B}(\mathcal{H})$ subset needed; φ_R does not map to density operators, observables,

B Level-4 Structural Inputs

Axioms K5–K7 and the bridge theorems reference four structural predicates supplied by the experimental-design layer (“Level 4”). They are collected here for self-containedness; none modifies K1–K8.

Shared validity demand D_{joint} . $D_{\text{joint}}(A, B, \text{Arch}) \in \{0, 1\}$ evaluates to 1 when the comparison architecture Arch (the experimental design) demands that A and B support one shared registration-validity claim. D_{joint} is a structural feature of Arch, not imposed by any single observer.

Joint-check predicate requires_K_joint . $\text{requires_K_joint}(A, B) = 1$ iff all of: (a) A and B are each valid or provisionally valid within their own K-side; (b) A, B are brought under a shared D_{joint} ; (c) D_{joint} requires both to be assessed as parts of the same registration target, history, counterfactual claim, or validity claim; (d) the truth of D_{joint} cannot be evaluated while leaving A, B in fully independent K-sides; (e) preserving D_{joint} requires a K_{joint} in which A, B are jointly valid. It is 0 iff no shared D_{joint} is imposed, or D_{joint} can be evaluated without embedding A, B into one candidate K_{joint} .

Comparison context C_K . C_K is the minimal shared frame in which two registration acts are evaluated for compatibility. It requires: (a) both acts admitted into the same comparison domain; (b) both indexed to the same registration target; (c) comparison does not presuppose joint validity. C_K is strictly weaker than AdmJoint : it enables comparison without requiring that joint validity be preserved.

Admissible joint space AdmJoint . $\text{AdmJoint}(K_{\text{joint}}; A, B) = 1$ iff there exist embeddings $i_A: A \rightarrow K_{\text{joint}}$ and $i_B: B \rightarrow K_{\text{joint}}$ such that:

- (i) the embeddings preserve act, outcome, certification, registration time/order, and validity;
- (ii) self-certification remains intrinsic to each embedded act;
- (iii) the six validity conditions below remain satisfied for each embedded structure;
- (iv) no required registration-state update in K_{joint} invalidates either embedded structure while both are still claimed as jointly valid;
- (v) K_{joint} does not import an external certifier as the source of self-certification.

The six *validity conditions* for a registered measurement are: physical occurrence; admission across the K-side boundary; process membership; self-certification ($\sigma_R(M) = 1$); default validity ($V = 1$); and non-invalidation (no later contradicting registration).

C Supporting Bridge Theorems: T2, T3, and AJVS

These results support T9 and the EWF application (Remark in §4). T2 and T3 are Class D; AJVS is a declared semantic postulate. None modifies K1–K8.

C.1 T2 — K-Incommensurability Derivation

Theorem C.1 ($\perp_K$ derivation — T2). *For K -spaces K_A, K_B : $K_A \perp_K K_B$ iff $\text{requires_K_joint}(A, B) = 1$ and no admissible K_{joint} exists satisfying AdmJoint (i)–(v) (Appendix B) while preserving K4 (default validity) and K5 (no invalidation) for both embedded sides simultaneously.*

Derivation (K5-conflict trace). Under a candidate K_{joint} , suppose $\exists k_A \in i_A(K_A), k_B \in i_B(K_B)$ with $k_B \perp k_A$ within C_K and $\text{Auth}(k_B \rightarrow k_A, C_K) = 1$ (K6). Then K5 forces $V_{\text{prov}}(k_A) \rightarrow 0$ or $V_{\text{prov}}(k_B) \rightarrow 0$, violating AdmJoint (iv) (no invalidation while both are jointly claimed valid). Hence no admissible K_{joint} exists, i.e. $K_A \perp_K K_B$. The K5 conflict is *sufficient*, not necessary: admissibility can also fail by a closure lock (K7: if K_A has already closed, no new act demanded by D_{joint} can be instantiated, so AdmJoint (i) fails with no K5 contradiction).

C.2 T3 — Extended Wigner’s Friend Bridge

Theorem C.2 (Bridge_EWF — T3). *In an Extended Wigner’s Friend configuration with laboratory time order $t_F < t_W$, where M_F registers a definite outcome o_F ($k_F = \langle M_F, o_F, 1, t_F, 1 \rangle$) and M_W registers the same laboratory as a coherent superposition in which no definite o_F is preserved ($k_W = \langle M_W, o_W, 1, t_W, 1 \rangle$): under a candidate K_{joint} , $k_W \perp k_F$ within C_K with $\text{Auth}(k_W \rightarrow k_F, C_K) = 1$, so K5 forces $V(k_F) \rightarrow 0$ or $V(k_W) \rightarrow 0$, violating AdmJoint (iv). Consequently $M_W \perp M_F$ at the act level.*

The derivation presupposes the EWF ordering $t_F < t_W$ (it is not stated for $t_W < t_F$) and is conditional on AJVS below.

C.3 AJVS — Axiom of Joint Validity Semantics

Postulate C.1 (AJVS). *A candidate K_{joint} satisfies $D_{\text{joint}}(A, B)$ iff it hosts the original K -side validity claims of both K_A and K_B as jointly evaluable first-order claims in a shared C_K . Hosting only meta-descriptions of the form “within K_A , M_A registered o_A ” does not satisfy D_{joint} : relativizing registration contents to sub-context descriptions abandons D_{joint} rather than satisfying it.*

AJVS is a declared semantic postulate, separate from K1–K8. If AJVS is rejected, the T3 conclusion does not follow from K1–K8 alone; K1–K8 remain structurally valid, and only the semantic scope of D_{joint} changes.

D N-Observer Colimit Existence (T4-H)

Let $\mathcal{C}_K$ be the category whose objects are K-spaces (sets satisfying K1–K8) and whose morphisms are K1–K8-preserving embeddings.

Theorem D.1 (Colimit existence — T4-H). *Every finite diagram D in $\mathcal{C}_K$ has a colimit $\text{colim}(D)$ in $\mathcal{C}_K$.*

Verification (4 steps).

S1 Category. $\mathcal{C}_K$ has identities, composition, and associativity of K1–K8-preserving embeddings.

S2 Construction. $K_{\text{colim}} = (\coprod_i K_i) / \sim$, with all five tuple fields well-defined: t assigned lexicographically and V as the embedding-time snapshot (K8); the order $<_{\text{colim}}$ is the transitive closure of the embedded orders plus cross-structure relations (T1-generalized).

S3 K1–K8 preservation. All eight axioms survive the quotient. Key lemma (T-PRES): K8-morphisms preserve t , so $t_{\text{colim}}([k, i]) := t_i(k)$ is representative-independent; K2 acyclicity follows, K5 cross-component $\perp$ is content-based, and the monotone V -dynamics are non-contradictory.

S4 Universal property. For any compatible cocone $\{f_i\}$, the map $u([k, i]) := f_i(k)$ is well-defined (diagram compatibility + T-PRES), K8-preserving, and unique.

Hence K_{colim} exists, satisfies K1–K8, and is universal. The construction is constructive for $N = 2$ (this is T1) and extends to all finite N .

E Associativity of K_{joint} (T5)

We prove T5 (§4) conditional on: (C1) T4-H (Appendix D); (C2) requires $\text{K_joint} = 1$ with AdmJoint satisfied for each pairwise and triple construction; (C3) the global commutativity condition F7d, established below.

Theorem E.1 (Associativity — T5, conditional). *Under (C1)–(C3): $K_{\text{joint}}(K_{\text{joint}}(A, B), C) \cong K_{\text{joint}}(A, B, C)$ as K1–K8-structured sets.*

Lemma B1 (field preservation). Any K8-preserving morphism preserves all five tuple fields as equalities; in particular $t_{\text{colim}}([k, i]) = t_i(k)$ is path-independent (T-PRES).

Lemma B2 (content-based K5). For $[k, i], [k', j]$ in the colimit with $i \neq j$, K5 fires iff (i) $i \neq j$ (observer identity), (ii) $o([k, i]) \neq o([k', j])$ (outcome content), and (iii) $t_{\text{colim}}([k, i]), t_{\text{colim}}([k', j])$ are co-temporal. All three depend only on tuple content, hence are path-independent.

Theorem B3 (F7d holds). By B1–B2, the set of K5 events firing against any k is identical along the incremental and one-shot construction paths; therefore V is path-independent and the global commutativity condition F7d holds.

Lemmas A1–A4 (universal property). $K_{\text{joint}}(A, B)$ and the iterate $K_{\text{joint}}(K_{\text{joint}}(A, B), C)$ are objects of $\mathcal{C}_K$ (T4-H Step 3). The composed embeddings j_A, j_B, j_C form a valid cocone for the 3-diagram $D = \{K_A, K_B, K_C\}$ (B3 gives V -agreement), and $K_{\text{joint}}(K_{\text{joint}}(A, B), C)$ satisfies the universal property for D : the T1 universal property gives a unique u_{AB} , and the T4-H universal property (Appendix D, S4) then gives a unique mediating u , with uniqueness inherited from both.

Proof of T5. Both $K_{\text{joint}}(K_{\text{joint}}(A, B), C)$ (Lemmas A1–A4) and $K_{\text{joint}}(A, B, C)$ (T4 + T4-H) are colimits of the same diagram D . Two colimits of one diagram are canonically isomorphic, so the unique mediating morphisms compose to identities in both directions, and the isomorphism is K1–K8-preserving. Thus K_{joint} is associative up to K1–K8-preserving isomorphism. T5 asserts K-side structural isomorphism only — not ρ -side associativity, physical outcome composition, or any modification of standard QM. $\square$